

Bell Pepper Price Analysis

A Data-Driven Framework for Agricultural Market Strategy

Project Overview



Transforming Raw Market Data into Actionable Insights

Core Objectives



Granular Analysis

Deep-dive tracking of market trends and specific price fluctuations across Red, Green, and Yellow Bell Peppers.



High-Rigor Pipeline

A systematic progression from initial data hygiene and complex feature engineering to advanced machine learning models.



Actionable Insights

Empowering the agricultural industry with interpretable, data-backed guidance for strategic decision-making and forecasting.

Data & Feature Taxonomy

Category	Extracted Features
Categorical	p_color
Numerical	vietnam_season, price, total_volume, brazil, india, vietnam, indonesia, china, jordan_max_price, jordan_min_price, demand, supply
Boolean	brazil_season, indonesia_season, india_season, china_season
Date	week_start_dt, week_end_dt



Noise Removal Systematic elimination of invalid records and rows containing artificial placeholder values (e.g., -1).



Missing Value Assessment Comprehensive programmatic audit mapping NA values across all categorical and numerical features.



Linear Imputation Application of linear interpolation to accurately resolve missing data points while maintaining time-series continuity.



Temporal & Schema Standardization Conversion of string dates to formal datetime objects, alongside automated verification of column data types.

Feature Engineering

Temporal Extraction

Designed to capture cyclical market behaviors and intrinsic seasonality.

- **Year:** Captures macro-economic annual shifts.
- **Month:** Identifies monthly agricultural cycles.
- **DayOfMonth:** Pinpoints granular intra-month volatility.

Advanced Statistical Windowing

Developed to capture historical momentum and price volatility over time.

- **Rolling:** Computes moving statistics over a fixed window.
- **Expanding:** Cumulative stats incorporating all past data.
- **EWM:** Prioritizes recent market shifts over older data.
- **EWMA:** Smooths fluctuations using specific weight spans.

Price Hierarchy by Color

Key EDA Insight: Across all monthly, yearly, and weekly resamples, a strict structural pricing hierarchy was identified within the dataset.



Yellow Peppers

Consistently exhibits the **highest mean price** across all observed timeframes, indicating premium market valuation.



Red Peppers

Follows as the **second most expensive** variety, maintaining stable mid-to-high tier pricing trends.



Green Peppers

Consistently maintains the **lowest mean price**, serving as the baseline commodity across all timeframes.

Statistical Testing

Correlation & Relationships

Validating relationships between continuous market drivers.

- **Pearson Correlation:** Validated via Bootstrap confidence intervals and structured via hierarchical clustering.
- **Partial Correlation:** Controlled for the **p_color** variable to isolate independent numerical relationships with price.
- **Categorical Association:** Leveraged Chi-Square, Cramer's V, and Theil's U to measure variable linkage strength.

Distribution & Variance (ANOVA)

Rigorous testing of distributional assumptions and price deltas.

- **Normality & Outliers:** Evaluated using Shapiro-Wilk testing, visual density plots, and defined anomaly thresholds.
- **Non-Parametric Tests:** Utilized Mann-Whitney U and Kruskal-Wallis to account for non-normal agricultural data distributions.
- **ANOVA Framework:** Executed post-hoc Tukey's HSD and power analyses to pinpoint significant price deltas between colors.

Dimensionality Reduction

Principal Component Analysis (PCA)

Managing high-dimensional feature sets following strict standardization.

- **Implementation:** Step-by-step extraction of components and eigenvalue spectrum mapping.
- **Quality Assessment:** Analyzed reconstruction error (original vs. reconstructed) to guarantee minimal information loss.
- **Visualization:** Generated 3D spatial mapping of the first three components to expose core data structures.

Non-Linear Manifold Learning

Overcoming the limitations of linear variance assumptions.

- **Market Reality:** Agricultural and market relationships inherently exhibit non-linear behaviors and hidden complexities.
- **Advanced Techniques:** Applied **t-SNE** and **UMAP** algorithms to visualize high-dimensional topology.
- **Cluster Discovery:** Successfully identified structural clusters that traditional linear methods (like PCA) often overlook.

Machine Learning Phase



Feature Selection

Utilized the **GreedyBorutaPy** algorithm to identify truly relevant predictors, effectively separating core signals from market noise in high-dimensional space.



Model Implementation

Transitioned from processed data to the systematic training and validation of the selected "best_model" tailored for accurate, robust price prediction.



Interpretability (SHAP)

Eliminated "black-box" outputs by applying SHAP to provide both **global** (feature ranking) and **local** (single prediction driver) interpretability.

Questions?

Thank you for exploring the Bell Pepper Price Analysis framework.

 Omar Soub

|  omars.soub@gmail.com

|  github.com/alsoub-labs